

EdgeSec: Design of an Edge Layer Security Service to Enhance IoT Security

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Abstract—With the widespread availability of connected smart devices, Internet of Things (IoT) is becoming the world's largest computing platform. These large-scale, heterogeneous and resource-constrained devices bring many significant new challenges to the design of efficient and reliable IoT systems. Security is one of the most crucial ones that need to be effectively addressed for the wide adoption of IoT systems. In this paper, we first present an in-depth analysis of security challenges in IoT. Then, we propose *EdgeSec*, the design of a novel security service which is deployed at the Edge layer to enhance the security of IoT systems. *EdgeSec* consists of seven major components that work together to systematically handle specific security challenges in IoT systems. Finally, the effectiveness of *EdgeSec* is demonstrated in the context of a typical IoT application, Smart Home.

Index Terms—Internet of Things, IoT, Security, Edge Computing, Smart Homes

I. INTRODUCTION

Internet of Things (IoT) is an emerging paradigm; it is considered as the third wave of innovation in information technology, after the Internet and mobile computing [1]. IoT encompasses everyday objects into the digital world by providing them computation, communication, and sensing capabilities, which collect valuable data. Intelligence can be extracted from the collected data to build a smarter world. According to CISCO, the number of devices connected to the Internet surpassed the world's population in 2009, and it is also predicted that there will be 50 billion devices connected to the Internet by 2020 [2]. McKinsey also reported that the annual economic impact of IoT will be between \$2.7 and \$6.2 trillion by 2025 [3]. The vision of IoT has promised to change the way we live, work, and learn.

IoT makes use of data collected from IoT things (end devices) to optimize the observation and control of the world in domains such as logistics, retail, military, healthcare, and so on [4]. However, IoT applications also bring many challenges with their benefits. Specifically, in IoT systems, the cyber and physical worlds are tightly coupled, which makes the physical world more vulnerable than before. Malicious instructions from the cyber world may cause severe damages to the physical world. If IoT security issues cannot be handled effectively, the disadvantages may outweigh the benefits [5]. Security has, therefore, been one of the key barriers that prevents the widespread adoption of IoT applications [6].

Designing efficient security services or mechanisms for IoT systems is challenging because of their special characteristics, including device resource constraints, heterogeneity of devices and communications, higher privacy requirements, trust managements, etc [7]. In addition, existing security issues in traditional Internet and enabling technologies such as Radio Frequency IDentification (RFID) and Wireless Sensor

Networks (WSN) further complicate the IoT security challenges [8]. Despite various efforts, security issues in IoT have not been addressed sufficiently and effectively.

In this paper, we propose a security service, *EdgeSec*, which is deployed at the Edge layer of the IoT architecture and aims to provide a systematic security solution for IoT systems. *EdgeSec* consists of seven major components, including Security Profile Manager (SPM), Security Analysis Module (SAM), Protocol Mapping (PM), Interface Manager (IM), Security Simulation Module (SSM), Request Handler (RH), and User Interface (UI). End devices get registered to *EdgeSec* with the help of SPM. For each end device, SPM creates a security profile and also collects the security requirements. Based on device security profiles and requirements, SAM decides where to deploy the security functions, i.e., at the Things layer, the Edge layer or the Cloud layer. Then the PM module chooses appropriate protocols to satisfy the security requirements based on the decision from SAM. Another component, IM, masks communication heterogeneity in IoT end devices. RH coordinates other components of *EdgeSec* and helps handle requests to access IoT resources in a secure way. Requests which may result in physical damages are tested in SSM before they are handled by end devices. Finally, UI is useful for administrators/users to interact with the *EdgeSec* components.

The main contribution of the paper is three-fold. First, we present an in-depth analysis of security challenges. Second, we propose the design of *EdgeSec*, a novel Edge layer security service, to provide a comprehensive solution for IoT security. Third, we present a case study of how to apply *EdgeSec* in a typical IoT application, Smart Home. To the best of our knowledge, this is the first work proposing a systematic security solution for IoT at the Edge Layer.

The remainder of the paper is organized as follows. In Section II, we present an overview of IoT and its applications. The security challenges and requirements of IoT systems are discussed in Section III. Section IV illustrates the detailed design of *EdgeSec*. A case study of *EdgeSec* application in Smart Home is provided in Section V. A set of related work is listed in Section VI. Finally, future work and conclusions are discussed in Section VII.

II. AN OVERVIEW OF IOT

Internet of Things (IoT) is becoming the next generation computing platform [4]. IoT aims to connect pervasive physical objects (Things) into an intelligent world, by making decisions that optimize the physical world based on mining data collected from the IoT things.

A. IoT Architecture

Understanding the IoT architecture is essential to build IoT systems. A typical three-layer architecture of IoT systems includes the Things layer, the Edge layer, and the Cloud layer [7]. The Things layer is mainly responsible for data collection and actuation to control the physical world. Most devices in the Things layer are resource constrained in terms of computational power, storage, and energy. Thus, they have to use various low-power, low-bandwidth communication protocols, and many of them cannot connect to the Internet directly. The Edge layer is introduced to help end devices. First, computation intensive tasks can be offloaded to edge devices. Second, edge layer can mask communication heterogeneity among end devices and connect them to the Internet. Third, edge devices help manage IoT end devices. The resource-rich Cloud layer is utilized not only to store, process, and analyze the collected data but also to provide additional support needed by many IoT applications.

B. Smart Home: A Typical IoT Application

IoT can be applied in many domains. Here we present a typical IoT application, Smart Home, which is becoming increasingly popular recently [9]. Gartner's IT Hype Cycle 2016 Report identified connected home as an emerging technology and predicted that a typical home could contain 500 or more smart devices by 2022 [10]. The vision of Smart Home is to add intelligence to everyday home objects, such as thermostats, door locks, surveillance cameras, garage doors, and so on. The added intelligence to physical objects offers many benefits to better human lives. These benefits include increased convenience, safety, security, and efficient usage of natural resources. For example, the Smart Home may adjust the blinds to save energy, automatically open the garage door when it senses the owner's vehicle approaching, or automatically order grocery when it detects low inventory.

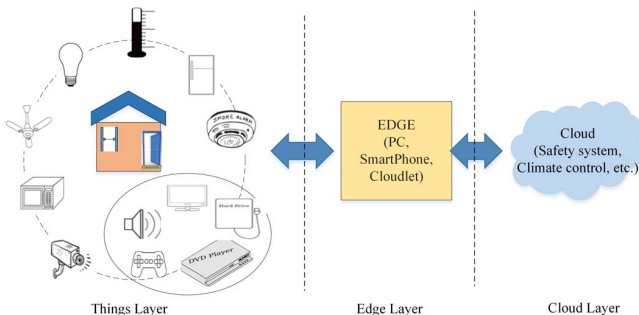


Fig. 1: The three-layer architecture of Smart Home.

Fig. 1 depicts the three-layer architecture of a Smart Home system. As seen in Fig. 1, end devices in the Things layer, such as thermostats, cameras, and smoke detectors, have ability to perceive the real world and to execute control instructions. It is also possible for some IoT things to form an ad hoc network. For example, an entertainment system ad hoc network can be formed by including smart TVs, DVD players, Net Drives, speakers, and gaming consoles. Other end devices are just

stand-alone devices, which are connected with the help of the Edge layer. Edge devices, such as smartphones, routers, smart meters, and home servers, also help connect end devices to the Internet. Finally, Smart Home data is collected and stored in Smart Home cloud, which provides additional support for IoT applications such as lighting control, climate control, safety system, and health monitoring.

III. SECURITY CHALLENGES AND REQUIREMENTS

To build an intelligent world based on IoT, we need to address many challenges, including standardization, interoperability, identity management, security, etc [11]. According to Wall Street Daily, security is the biggest barrier for the real-world IoT implementation, followed by privacy and connectivity issues [6]. Many security researchers and hackers are able to find vulnerabilities in IoT products. For instance, security researchers from the University of Michigan exploited vulnerabilities in Samsung Smart Home kit and demonstrated how they unlocked the doors of a home [12]. Miller and Valasek demonstrated how to disable the transmission and brake controls in a Jeep Cherokee [13]. Recently, a Distributed Denial of Service (DDoS) attack on DNS provider Dyn exploited vulnerabilities in millions of connected IoT devices [14]. There are many other cases where hackers compromised weak devices, such as toasters and freezers, and used them to control the entire home network. These cases exemplify the desperate needs of enhancing security in IoT systems. Society will not adopt the IoT systems widely until the security issues are addressed. This paper aims to propose a novel and comprehensive security solution for IoT systems.

The inherent characteristics of IoT make it challenging to secure IoT. First, end devices are resource constrained in terms of computational capability, power, and storage. For example, IoT applications can use devices with resources as low as 64 bytes of RAM, 2KB of storage, and 4-bit CPU architectures [15]. Therefore, conventional security mechanisms such as asymmetric key encryption based protocols are not applicable, as they need lots of resources including energy [16]. Furthermore, end devices are heterogeneous in nature. A single IoT application may use devices with various hardware capabilities running on different operating systems communicating with different protocols. It is very difficult to build general security solutions for heterogeneous devices. For example, an antivirus software can be installed on a PC or a smartphone but not on a smart bulb or a temperature sensor.

Additionally, end devices, in contrast to traditional computing devices, are typically deployed in unattended environments; they are therefore more prone to physical attacks. Having physical access to devices allows an adversary to extract information such as encryption keys, usernames, and passwords easier than before. Additionally, the sheer number of IoT devices exacerbate the security problems. Because hackers continuously exploit new vulnerabilities of IoT systems, we need to continuously update the firmware and add security patches. The devices may also have to be replaced to support new technologies developed for IoT systems. These

updates involve huge costs and technical difficulties. Moreover, the inclusion of new technology may also result in new security challenges. For example, RFID technology brings new security issues such as tag spoofing, buffer overflow, and unauthorized tag reading [8]. Addressing privacy issues in IoT systems is also challenging, because IoT systems involve a huge amount of personal data, especially when they are applied in healthcare applications. Having a strong identity is useful in protecting the security, but meanwhile it threatens the privacy of individuals. Therefore, it is very difficult to achieve a high level of both privacy and security. Finally, establishing trust management in IoT systems is also very difficult because of the high mobility nature of IoT devices. All of the above IoT characteristics make the design of security mechanisms extremely challenging.

Although securing IoT systems poses more challenges than securing conventional computing systems, all conventional security requirements are still applicable to IoT systems, including Confidentiality (Conf.), Data Integrity (DI), Availability (Avail.), Authentication (Auth.), and Privacy. For example, in a Smart Home application where smart cameras continuously take footage for security monitoring purposes, we need to make sure that the data remains confidential to preserve privacy. Also, the data should not be altered or compromised. Furthermore, the service should be available all the time to authorized users through effective authentication mechanisms.

IV. EDGESEC: AN EDGE LAYER SECURITY SERVICE

To address the IoT security challenges discussed above, we propose *EdgeSec*, an Edge layer security service to enhance the security of IoT systems. There are several advantages of exploiting the Edge layer to support security goals. First, the Edge layer has much more resources than the Things layer. End devices can leverage these resources to offload computation-intensive tasks, such as data encryption, key generation, and intrusion detection, to the Edge layer. Second, the Edge layer is close to IoT things and has direct or light multihop communication with them. This reduces the communication cost significantly for low capable IoT devices, and meanwhile achieves better real-time performance of IoT applications. Third, the Edge layer has more information than end devices about the whole system; therefore, it is possible to deploy optimized algorithms at the Edge layer. Fourth, edge devices can have a relatively stable relationship with the IoT things, which is very beneficial to establish trust between the IoT things and edge devices. Fifth, the Edge layer can be used to safeguard the privacy of individuals by enabling more user control over their personal information, or by applying secure aggregation algorithms. Sixth, the Edge layer has high-speed connection with the Cloud layer and it is easy for them to get extra support from Cloud layer services. Finally, the Edge layer is flexible enough to support various services needed by IoT things. Because of all those benefits, we propose to deploy the IoT security management at the Edge layer.

As seen in Fig. 2, the design of *EdgeSec* consists of seven major components as follows: Security Profile Manager

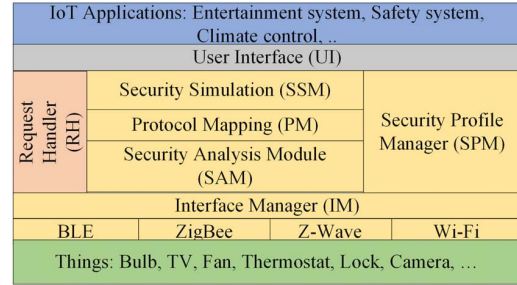


Fig. 2: The design of *EdgeSec*.

(SPM), Security Analysis Module (SAM), Protocol Mapping (PM), Interface Manager (IM), Security Simulation Module (SSM), Request Handler (RH), and User Interface (UI). SPM registers IoT things to *EdgeSec*. It creates a security profile and also collects security requirements of each end device. Based on device security profile and requirements, SAM decides where to deploy the security functions, i.e., whether at the Things layer, the Edge layer, or the Cloud layer. Then the PM module chooses appropriate protocols to satisfy the security requirements based on the decisions from SAM. IM is designed to mask communication heterogeneity in IoT things. Requests which may cause physical damages are tested in SSM before they are handled by IoT things. Coordinating other *EdgeSec* components, the RH module helps handle requests to access IoT resources in a secure way. Finally, interfaces are provided for administrators and users to interact with *EdgeSec* components through UI. In the rest of this section, we introduce each component in detail.

A. Security Profile Manager (SPM)

Analyzing and understanding the capabilities of the IoT end devices are essential to the provision of effective security solutions for them. The SPM module creates security profiles of end devices, including information such as computation power, storage size, energy source, the type of data collected, and the type of actions. These security profiles are essential to extract security requirements. Each IoT device tends to have its own specific security requirements, based on its intended use and context. For example, for a temperature sensor in a public park, confidentiality may not be a concern but availability is expected. The SPM module is designed to accomplish two tasks, device registration and profile creation.

1) *Device registration*: End devices need to get registered in order to be a part of the IoT system. Registration also helps establish trust between end devices and the Edge layer. Device registration can be completed in two ways. First, an end device can actively send a request to join the IoT system. The message will be forwarded to the Edge layer by devices already in the IoT systems. The closest edge device then authenticates that device and sends back registration acceptance message to complete the registration. Second, when the edge device receives a request to query or control the end device in the area covered by the edge device, it will broadcast a registration invitation message to initialize the device registration process.

TABLE I: Device categories: definitions and examples

Category	RAM	ROM	Power source	Security mechanisms supported	Communication protocols	Example devices
Constrained	Up to 10KB	Up to 128KB	Battery	Do not support any traditional security mechanisms	IEEE 802.15.4	Phillips Hue
Limited	10-32KB	128-512KB	Battery	Support symmetric key based protocols	IEEE 802.15.1	August smart lock
Restricted	32-128KB	512KB-10MB	Battery/AC power	Support symmetric and lightweight asymmetric key protocols	IEEE 802.15.4/ 802.11	Nest smoke detector
Normal	128KB and above	10MB and above	AC power	Supports all traditional security protocols	IEEE 802.11/ 802.15.1/ 802.15.4	Nest Thermostat, Samsung Smart TV

The registration will be completed after the end device receives and responds the invitation message.

2) *Profile creation*: At the end of the device registration, a security profile is created for the registered device. The security profile consists of (1) the category of the device in terms of computation power, storage size, and power source; (2) type of collected data; (3) type of actions it executes. We have categorized end devices into four categories as shown in Table I, based on physical features of these devices. For each type of devices, we also give examples of representative devices. *Constrained* devices are the weakest type of devices in the IoT ecosystem and are powered by batteries; they do not support any conventional security mechanisms. *Limited* devices can support some symmetric key based protocols but not asymmetric key based protocols. While *Restricted* and *Normal* devices are more powerful devices that can work with lightweight asymmetric key based protocols and all traditional security protocols respectively. Based on who has access to the data, data can be categorized into three classes, *Public*, *Internal*, and *Confidential*. Public data can be accessed by anyone. Internal data is available to entities of the same ad hoc network. Confidential data, however, can only be accessed after permission is granted by the owner of the device. In addition, three types of actions taken by IoT things are also defined based on potential consequences of these actions, including *critical* actions like unlocking the door, *risky* actions like switching off the power, and *controlled* actions like turning off the light. The highest level of security is needed to secure confidential data, followed by internal and public data. Similarly, the highest level of security should be provided to protect critical actions, followed by risky and controlled actions.

TABLE II: Examples of device security profiles

Devices	Category	Type of data	Type of action
Smart TV	Normal	Internal	Risky
Smart Bulb	Constrained	Internal	Controlled
Smart Lock	Limited	Confidential	Critical
Camera	Normal	Confidential	Controlled
Thermostat	Normal	Internal	Risky

Table II shows examples of security profiles of some Smart Home devices. The security profile of a smart lock indicates that the device has *limited* resources, it collects *confidential* data, and it executes *critical* actions. Since most Smart Home devices collect internal or confidential data, all the security requirements mentioned in Section III need to be satisfied for these Smart Home devices. In comparison, devices used in some other IoT applications (such as Smart City) may not

demand all security requirements; for instance, confidentiality is not required for temperature sensors in a public park.

B. Security Analysis Module (SAM)

The security analysis module is one of the core function components of EdgeSec. It implements two critical functions. One is to decide where to deploy the security functions to satisfy the various security requirements extracted from device profiles, i.e., at the Things layer, the Edge layer, or the Cloud layer. The other is to analyze the transitive access granting issue among the ad hoc networked devices. The transitive access granting is defined as getting the permission to access one end device may grant access to other devices in the same ad hoc network. It may introduce many security issues as discussed in [13]. The Edge layer needs to detect the transitive access granting and judge whether it should be allowed or not. The details of the two functions are explained as follows.

1) *Security functions deployment*: To address security challenges resulted from the resource-constrained end devices, it is necessary to offload some security functions to the Edge layer or the Cloud layer. A critical decision is how to distribute the deployment of the security functions. The decision is made based on the category of the devices and security requirements extracted from the type of data and actions. To facilitate the process of the decision making, a decision rule library needs to be created. For example, one rule dictates that the security functions to satisfy security requirements for *constrained* end devices should be deployed at the Edge layer. The rule library is maintained by the administrator of the system, which can be either the owner of the devices or a trusted third party.

Table III displays examples of security function deployment decisions for some Smart Home devices. We can observe that for constrained devices like a smart bulb, all security functions are deployed at the Edge layer. For normal end devices like smart TV, security functions for confidentiality, data integrity, and authentication are deployed to these end devices, while security functions for availability and privacy are deployed at the Edge layer. This is more effective because installing an Intrusion Detection System (IDS) and deploying privacy-preserving algorithms require more information as input to IDS or algorithms.

TABLE III: Examples of security functions deployment

Device	Conf.	DI	Auth.	Avail.	Privacy
Smart TV	End	End	End	Edge	Edge
Smart Bulb	Edge	Edge	Edge	Edge	Edge
Smart Lock	Edge	Edge	Edge	Edge	Edge
Camera	End	End	End	Edge	Edge
Thermostat	End	End	End	Edge	Edge

2) *Security Dependency Analysis*: It is known that in IoT ecosystem, a set of end devices may be linked to form a cluster, which work together as an ad hoc network. Fig. 1 contains an example of a home entertainment system in the format of an ad hoc network, which consists of devices such as smart TVs, DVD players, speakers, Gaming consoles, and Net Drives. Gaining access to one device such as smart TV will extend access to other devices such as speakers and DVD player, such a transitive access granting may cause severe security problems. The worst consequence can be the whole IoT system is compromised because of the compromise of one device. Therefore, the transitive access granting among these devices need to be analyzed and modeled.

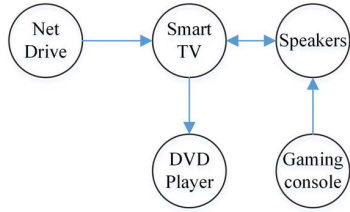


Fig. 3: Transitive access granting shown as a directed graph.

The directed graph is used to model the transitive access granting among the multiple end devices in an ad hoc network. In a directed graph, a set of nodes are connected by edges that direct from one node to another. Nodes denote devices and edges represent the transitive access granting. Fig. 3 shows an example of modeled transitive access granting within an entertainment ad hoc network using a directed graph. As shown in the figure, having access to the smart TV can extend the access to the DVD player and the speakers. Using the directed graph modeling the transitive access granting among devices, the risk of granting permission to access one device (device A) needs to be carefully evaluated by examining the set of devices with transitive access granted. These devices can be identified by traversing the graph with device A as the starting point. For example, traversing the graph from the Net Drive, one will gain access to the smart TV, the DVD player, and the speakers. If the risk evaluation reveals potential security issues due to transitive access granting, either the devices in the transitive access list will be further contacted for access permissions or the requested device will be notified not to extend the access granting.

C. Protocol Mapping (PM)

After SAM make decisions on where to deploy the security functions, protocols are needed to support these decisions. The PM module selects appropriate security protocols based on the available resources at the devices to satisfy the security requirements specified in the device's security profile. Here we assume that a set of protocols are already in the protocol library and there is an appropriate protocol to satisfy the identified security requirements and resource constraints. For example, lightweight Datagram Transport Layer Security (DTLS) based protocols can be utilized for resource-limited devices [17]. How to design these protocols is beyond the

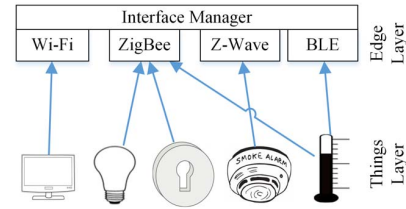


Fig. 4: Interface Manager.

scope of this paper. If there are several candidate protocols, the lightweight one will be selected. Based on security functions deployment decisions, it is possible that the PM module needs to choose different protocols for different communication segments. In other words, if SAM decides to deploy security functions at the Edge layer, then PM will have to choose a set of protocols for both the cloud-to-edge communication segment and the edge-to-end communication segment. Novel security protocols need to be explored to secure both communication segments, especially the one between the Edge layer and the Things layer.

Table IV demonstrates examples of selected protocols by PM for devices in Smart Home application. As we can observe in the table, for each device, PM selects appropriate security protocols in order to satisfy specific security requirements as introduced in Section III. For example, security functions for smart bulb are deployed at the Edge layer. Transport Layer Security (TLS) protocol is used to preserve confidentiality, data integrity, and authentication for cloud-to-edge communication. Firewalls and IDS systems are deployed at the Edge layer for preserving availability. To preserve the privacy, k-anonymity privacy preserving protocol [18] is utilized. The smart bulb cannot support security protocols for edge-to-end communication as it does not have enough resources.

D. Interface Manager (IM)

One of the most significant challenges in IoT security is interconnectivity. It is caused mainly by the heterogeneity of communication protocols used by end devices. IM is designed to handle this challenge. It first detects the type of the communication protocol used by end devices. Then it forwards the package to the Edge layer device that supports the detected communication protocol. For example, the Edge layer consists of devices that support different communication protocols, such as Wi-Fi, Bluetooth Low Energy (BLE), ZigBee, Z-Wave, and WirelessHART. Messages from the end device can be handled by the corresponding edge device that supports the same protocol as the end device. Fig. 4 depicts how IM masks the communication heterogeneity of end devices.

E. Security Simulation Module (SSM)

One characteristic of IoT systems is the tight coupling between the physical and cyber worlds; instruction from the cyber world may have a significant impact on the physical world. For example, turning off the A/C system in a data center may lead to system crash due to overheating. It will result in significant direct and indirect financial loss. Therefore,

TABLE IV: Examples of protocol mapping

Device	Conf.	DI	Auth.	Avail.	Privacy
Smart TV	TLS	TLS	TLS	Firewall and IDS	k-anonymity
Thermostat	TLS	TLS	TLS	Firewall and IDS	k-anonymity
Smart Lock	TLS - for cloud to edge; [19] - for edge to end	TLS - for cloud to edge; [19] - for edge to end	TLS - for cloud to Edge; [19] - for edge to end	Firewall and IDS	k-anonymity
Smart Camera	TLS	TLS	TLS	Firewall and IDS	k-anonymity
Smart Bulb	TLS - for cloud to edge; Not supported - for edge to end	TLS - for cloud to edge; Not supported - for edge to end	TLS- for cloud to edge; Not supported - for edge to end	Firewall and IDS	k-anonymity

actuation instructions from the cyber side of the system should be evaluated carefully to reduce potential loss before they are executed by the physical world. SSM aims to evaluate the consequence of critical instructions to the physical world.

This module maintains a list of critical instructions. Whenever a request is classified to be critical, the instruction will not be forwarded to the actuator directly; instead, a simulator that models the status of the system will be used to execute the instruction and to simulate the evolution of the system. If any severe consequence is detected, the instruction will be blocked. In the case of no severe consequence, the instruction will be forwarded. Otherwise, if the result of the simulation cannot decide whether or not to forward the request, the case can be escalated to the administrator of the device, who will make a final decision.

F. Request Handler (RH)

A big portion of data communication happens within the ad hoc peer-to-peer network built by end devices. Because capabilities of end devices vary, it is difficult to satisfy security requirements in this type of communications. Therefore, the Edge layer is utilized to satisfy the security requirements. In addition, a component is needed to coordinate multiple above defined components of EdgeSec and to handle the requests from devices not a part of the ad hoc network. The RH module is designed to fulfill this gap. To control the overhead to the Edge layer, we only make the Edge layer get involved in the connection establishment phase of above communications, when a request to access data is sent to one end device from devices within or outside of the ad hoc network.

With the help of the Edge layer, the request can be handled as follows. After any request that is either coming from external or internal applications has been received by RH, it first contacts SAM to check whether the request is to a stand-alone device or to a device in an ad hoc network. If the request is to a device in an ad hoc network, the SAM traverses through the directed graph to get the list of transitive access granted devices. It analyzes the potential security issues by checking the security profiles of the devices to make a decision. Second, RH searches the incoming request in the critical request list. If the request is critical, SSM will be used to evaluate the consequence of the request. If the outcome has severe effects on the physical world, the request will be blocked; otherwise, the request will be forwarded to the end device.

G. User Interface (UI)

The UI module helps users or administrators to interact with different components of EdgeSec. It handles most configuration issues in EdgeSec. For example, using the interfaces in UI,

administrators can configure the rule library in SAM. The list of critical actions in SSM can be updated according to specific application requirements. Additionally, the escalated critical requests from SSM module can be handled by administrators using UI as well.

H. Discussion

Previous sections present the detailed design of EdgeSec, which aims to address many challenges brought by special characteristics of IoT systems. The challenge brought by resource constraints of IoT things is addressed by offloading the computationally intensive tasks to the Edge layer. SPM analyzes the capabilities and security requirements of each end device. Based on the capabilities and security requirements, SAM module distributes the security functions. SAM has also addressed the transitive access granting problem, which causes many of the significant security issues that are reported recently [14]. Communication heterogeneity issues in IoT systems are addressed by IM, which masks the communication heterogeneity to higher layers. The potential dangerous impact of IoT systems on the physical world is evaluated by SSM through assessing the possible consequences of an actuation request. The challenges caused by a multitude of devices in IoT can be addressed using multiple edge devices in the Edge layer, which forms a peer-to-peer ad hoc network to manage a set of large-scale IoT devices. The design of EdgeSec also helps address the trust management problems by using the Edge layer to manage the IoT end devices. As the Things layer and the Edge layer are close to each other, edge devices can have a relatively stable relationship with end devices. Having information of multiple end devices available at the Edge layer, we can easily exploit several privacy-preserving algorithms to preserve privacy. Overall, the design of EdgeSec addresses most of the significant challenges that concern IoT security.

V. A CASE STUDY: EDGESEC FOR SMART HOME

In this section, an application of EdgeSec in a typical IoT application, Smart Home, is presented. As shown in Fig. 1, there are a set of end devices in Smart Home, including smart thermostats, smart TVs, and so on. Among them, smart bulbs and smart fans are constrained devices; smart locks and smart ovens are limited devices; smoke detectors and smart refrigerators are restricted devices; smart TVs, smart cameras, and smart thermostats are normal devices. There is also an ad hoc network that consists of devices for entertainment systems including smart TVs, speakers, DVD players, Net Drives, and Gaming consoles. These end devices make use of various communication technologies. For example, smart TVs

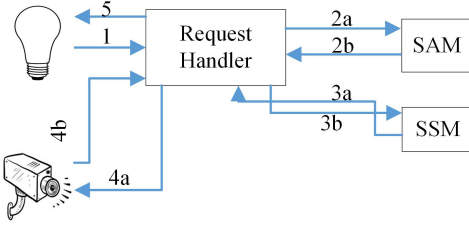


Fig. 5: Handling a local request.

and cameras support well-known Wi-Fi protocol; smart bulbs uses ZigBee protocol; smart locks supports BLE; and smart thermostats supports Wi-Fi, ZigBee, and BLE protocols.

To be a part of the Smart Home system, end devices need to register to EdgeSec, either via sending registration request directly to edge devices or by accepting the broadcast invitation message from edge devices. When the end devices are registered, a security profile is created for each of the registered devices. Table II shows examples of security profiles of registered devices. Based on the security profiles, SAM decides where to deploy security functions to satisfy devices' various security requirements as shown in Table III. Afterward, PM searches the protocol library to find appropriate protocols to support these security functions. The selected security protocols are displayed in Table IV. Completing these steps, we have established an EdgeSec supported Smart Home system. Next, we present several typical scenarios to illustrate further how EdgeSec works. These scenarios illustrate several typical cases of security concerned communications in Smart Home.

A. Scenario#1: Handling a local request

In this scenario, a smart bulb requests access to smart cameras that has room occupancy information. This is a local request as both devices are on the same network. Fig. 5 shows the steps of how EdgeSec handles the request from the smart bulb. After the RH module receives the request from the smart bulb at step 1, it contacts SAM to check whether the request is to access a stand-alone device or a device in an ad hoc network at step 2a. SAM responds and indicates that it is a stand-alone device at step 2b. In step 3, RH contacts SSM for criticality check, and SSM responds that it is not a critical request. Then RH forwards the request to the smart camera at step 4a, and it responds that room is vacant (step 4b). After RH forwards the response from the camera to the smart bulb, the smart bulb will be turned off. All the communication are handled by the security protocols selected by PM.

B. Scenario#2: Handling a cloud request

In this scenario, the smart lock application that resides in the cloud, requests access to the smart lock to unlock the door. Fig. 6 shows the steps of how EdgeSec handles the request. After RH receives the request from the smart lock application at step 1, it contacts SAM to check whether the request is to access a stand-alone device or a device in ad hoc network at step 2a. SAM responds and indicates that it is a stand-alone device (step 2b). Next step, RH contacts SSM for criticality check and SSM responds that it is not a critical request. RH

forwards the request to the smart lock, and the smart lock unlocks the door and responds with a message that door is unlocked at step 4. Finally, RH will forward this response to smart lock application.

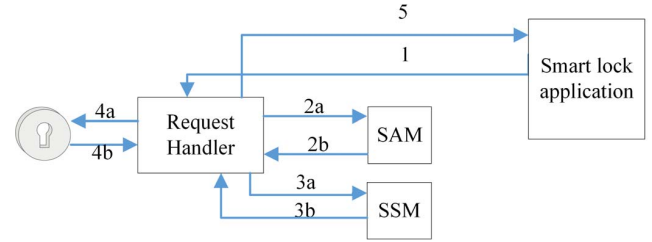


Fig. 6: Handling a cloud request.

C. Scenario#3: Handling a request to a device in an ad hoc network from the cloud

In this case, a video streaming application that resides in the cloud, requests access to the smart TV. As discussed in above scenarios that RH contacts SAM, SAM finds out that the request is to access a device in an ad hoc network. SAM traverses through the directed graph using the smart TV as the starting point and finds out that DVD players and speakers are the additional devices that this request can access. Then RH checks the security profiles of these devices and decides to forward the request. Next RH contacts the SSM, and it confirms that the request is not critical. The rest of the procedure is similar to Scenario#2.

VI. RELATED WORK

Several researchers have proposed End-to-End security protocols that securely connect end devices to the Internet. DTLS [20], HIP DEX [21], and minimal IKEv2 [22] are some examples. These protocols use asymmetric key based encryption algorithms. However, they are not applicable for many IoT devices as they require lots of computational resources. The work presented in [23] proposes a two-way authentication protocol based on DTLS, which uses X.509 certificates and RSA public key algorithms. Another similar approach [17] proposes to compress the DTLS header by leveraging the 6LoWPAN implementation techniques. In all of the above approaches, the capabilities of end devices used in the evaluation are still high. As the public key encryption is resource demanding and not suitable for constrained devices, some researchers propose to offload the resource demanding public key encryption to neighbor devices with enough resources. One such approach is proposed in [24]. This approach proposes to offload key establishment process to neighboring node with enough resources by assuming that the node is trusted. In another similar approach [25], authors propose an authentication and key agreement protocol assuming that each sensor node should have a pair-wise secret key with neighbouring sensor nodes.

The other approach to secure end devices is to offload security functions to gateways or edge devices. The work presented in [26] proposes to secure the data communication

in two segments, IoT things to the gateway and the gateway to the Internet. The data communication from the gateway to the Internet is secured by TLS protocol, and Symmetric key encryption protocols are used for the gateway to IoT things communication. A similar work is presented in [19]. The proposed work is to secure isolated smart grid devices by using the smart grid reader as a bridge connecting the cloud and isolated smart grid devices. This approach devises two secure protocols, one for the reader-cloud communication and the other for the reader-device communication. In [27], authors propose to offload authentication and access control to either the cloud or gateways. The decision of offloading is made based on the resources that gateway device possesses. Montero *et al.* [28] also propose an architecture to secure end user devices at the network edge. The approach is user-centric and promises to provide a uniform security solution for all the devices used by a single user by creating virtual domains for each device at the edge of the network.

VII. CONCLUSIONS AND FUTURE WORK

Security is the biggest concern for IoT systems. It is very difficult to address the security challenges of IoT systems because of its new characteristics. In this paper, we first analyze various challenges that make it difficult to secure IoT systems. Then we design EdgeSec, an Edge layer security service, which enhances IoT systems security. The seven major components of EdgeSec are discussed in detail. Furthermore, EdgeSec is applied in Smart Home application in the context of several use cases.

We plan to extend our work in three perspectives. First, the prototype of EdgeSec will be implemented by exploiting Software Defined Network (SDN) and Virtualization technologies. Second, a Smart Home application will be built to test and evaluate the efficiency and effectiveness of the implemented prototype system. Third, we will investigate next-generation light-weight security algorithms and protocols that can be used in our prototype to enhance the IoT security. Finally, how to organize edge devices into a secure and efficient edge layer will be explored.

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